MTH2008 Real Analysis: Exercise Sheet 1

This Exercise Sheet covers the material presented in Week 1. By completing the exercises, you should develop an understanding of axioms that define the real number, sets and set operations, and the topology of the real line.

1. In each of the following cases determine the supremum and the infimum of S, justifying your answer.

(i) $S = \left\{ 1 + \dfrac{(-1)^n}{n} \,\middle|\, n \in \mathbb{N} \right\}$

(ii) $S = \{e^n \,|\, n \in \mathbb{N} \cup \{0\}\}$

Solution:

(i) Observe that if n is odd, then $1+(-1)^n n^{-1} = 1-n^{-1}$, while if n is even $1+(-1)^n n^{-1} = 1+n^{-1}$. Hence $x \leq 1 + 1/2$ for every $x \in S$. We therefore have that $\sup S = 3/2$. Indeed if $v < 3/2$, then there exists $x \in S$ such that $x > v$, in particular with $x = 3/2 = 1 + 1/2$.

We also claim that $\inf(S) = 0$. Indeed, $0 \leq 1 - 1/n \leq 1 + 1/n$ for all integers $n \geq 1$. Further, if $v > 0$, then we can find $x \in S$ such that $x < v$, as we can take $x = 0 = 1 - 1/1$.

(ii) The set S is unbounded from above. Indeed, for every $M > 0$, we can find an $x \in S$ such that $x > M$. For instance, one may choose $n > \ln(M)$, in which case $x = e^n > M$. On the other hand, as the exponential function is an increasing function and as $1 = e^0$ for every $x \in S$, we have that $1 < x$. Further, if $v > 1$, then we can find $x \in S$ such that $x > v$, as we can take $x = 1 = e^0$. Therefore, we have that $\inf(S) = 1$.

2. Let X be a set of positive real numbers with supremum $\alpha = \sup X \in \mathbb{R}$. Let

$$Y = \{x^2 + \sqrt{3} \,|\, x \in X\}$$

determine the supremum of Y. Justify your answer.

Solution: We claim that $\sup Y = \alpha^2 + \sqrt{3}$. Since $\alpha = \sup X$, then $x \leq \alpha$ for every $x \in X$, hence $x^2 + \sqrt{3} \leq \alpha^2 + \sqrt{3}$. Now suppose that $\sup Y = \beta < \alpha^2 + \sqrt{3}$, then $x^2 + \sqrt{3} \leq \beta$ for all $x \in X$, or in other words,

$$x \leq \sqrt{\beta - \sqrt{3}},$$

for all $x \in X$. However, since $\beta < \alpha^2 + \sqrt{3}$, this implies that

$$x \leq \sqrt{\beta - \sqrt{3}} < \sqrt{\alpha^2 + \sqrt{3} - \sqrt{3}} = \alpha.$$

for all $x \in X$, contradicting the fact that $\alpha = \sup X$.

3. Prove the following statements.

(i) For each $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ such that $x < n$.

(ii) For all $x, y \in \mathbb{R}$, if $x > 0$, then there exists $n \in \mathbb{N}$ such that $y < nx$.

(iii) For all $x \in \mathbb{R}$, if $x > 0$, then there exists $n \in \mathbb{N}$ with $0 < 1/n < x$.

Solution:

(i) Let $x \in \mathbb{R}$ and suppose, by way of contradiction, that $n \leq x$ for all $n \in \mathbb{N}$. This would imply that the set of natural numbers is bounded above in $\mathbb{R}$, and thus by the completeness axiom has a supremum $\beta \in \mathbb{R}$. Since $\beta - 1 < \beta$, there exists $m \in \mathbb{N}$ such that $\beta - 1 < m$. Therefore, $\beta < m + 1$, but $m + 1$ is an integer, contradicting the fact that β is the supremum, and hence an upper bound, of $\mathbb{N}$.

(ii) Let $x, y \in \mathbb{R}$ with $x > 0$. Consider the real number y/x. By Part (i) there exists $n \in \mathbb{N}$ such that $y/x < n$. Therefore, we may conclude that $y < nx$.

(iii) Let $x \in \mathbb{R}$ with $x > 0$. From Part (ii), taking $y = 1$, we conclude that there exists $n \in \mathbb{N}$ with $0 < 1 < nx$, or in other words, that there exists $n \in \mathbb{N}$ with $0 < 1/x < n$.

4. Show that $\inf S \leq \sup S$ for any bounded nonempty set S of real numbers, and give necessary and sufficient conditions for equality.

Solution: Let S be a bounded nonempty set of real numbers, and set $m = \inf S$ and $M = \sup S$. Since M is an upper bound for S and since m is a lower bound for S, every $s \in S$ satisfies $m \leq s \leq M$. Therefore, we have that $m \leq M$.

We claim that $\inf S = \sup S$ if and only if $S = \{c\}$ for some $c \in \mathbb{R}$, that is, S is a singleton. To see that let us suppose that $\inf S = \sup S = c$. In which case, for any $s \in S$, we have that $c = \inf S \leq s \leq \sup S = c$, and hence $s = c$. Thus, every element of S equals c, and so $S = \{c\}$. On the other hand, if $S = \{c\}$, then the greatest lower bound and least upper bound of S are both c, and hence $\inf S = \sup S = c$.

5. Let S and T be non-empty sets of real numbers and define

$$S + T = \{s + t \mid s \in S \text{ and } t \in T\}.$$

Show that

$$\sup(S + T) = \sup S + \sup T$$

if S and T are bounded above.

Solution: Assume S and T are bounded above and non-empty, and set

$$M = \sup S \quad \text{and} \quad N = \sup T.$$

Observe that these exist by the completeness axiom, and note that since S and T are non-empty, there exists $s \in S$ and $t \in T$, and so $S + T$ is non-empty since $s + t \in S + T$. If $x \in S + T$, then $x = s + t$ for some $s \in S$ and $t \in T$. Since, $s \leq M$ and $t \leq N$, we have that $x = s + t \leq M + N$. Hence $M + N$ is an upper bound of $S + T$, and so $\sup(S + T)$ exists and $\sup(S + T) \leq M + N$. To prove that this is in fact an equality, let $\epsilon > 0$. In which case, there exist $s \in S$ and $t \in T$ with

$$s > M - \epsilon/2, \quad \text{and} \quad t > N - \epsilon/2.$$

This however implies that

$$s + t > (M - \epsilon/2) + (N - \epsilon/2) = M + N - \epsilon.$$

In other words, $S + T$ contains an element exceeding $M + N - \varepsilon$. Since this holds for every $\epsilon > 0$, we have that $M + N$ is the least upper bound of $S + T$, namely $\sup(S + T) = M + N = \sup S + \sup T$.

6. Let $\mathcal{F}$ be a collection of subsets of $\mathbb{R}$ and define

$$I = \cap\{F \mid F \in \mathcal{F}\} \quad \text{and} \quad U = \cup\{F \mid F \in \mathcal{F}\}.$$

Prove that

(i) $I^c = \cup\{F^c \mid F \in \mathcal{F}\}$, and

(ii) $U^c = \cap\{F^c \mid F \in \mathcal{F}\}$.

These two equalities are sometimes called *De Morgan's law*.

Solution:

(i) If $x \in I^c$, then $x \in \mathbb{R}$ and $x \notin I = \cap\{F \mid F \in \mathcal{F}\}$. This means that there exists an $F \in \mathcal{F}$ such that $x \notin F$, or in other words, there exists an $F \in \mathcal{F}$ such that $x \in F^c$. This implies that $x \in \cup\{F^c \mid F \in \mathcal{F}\}$. Thus, we have shown that $I^c \subseteq \cup\{F^c | F \in \mathcal{F}\}$.

To show the required equality, we prove $\cup\{F^c | F \in \mathcal{F}\} \subseteq I^c$. To this end, let $x \in \cup\{F^c | F \in \mathcal{F}\}$. This implies that there exists an $F \in \mathcal{F}$ such that $x \in F^c$, or in other words, there exists an $F \in \mathcal{F}$ such that $x \notin F$. Hence, $x \notin I = \bigcap\{F \mid F \in \mathcal{F}\}$, namely, $x \in I^c$. Thus, we have that $\cup\{F^c | F \in \mathcal{F}\} \subseteq I^c$ as required.

(ii) If $x \in U^c$, then $x \in \mathbb{R}$ and $x \notin U = \cup\{F \mid F \in \mathcal{F}\}$. This means that $x \notin F$ for every $F \in \mathcal{F}$. Hence, $x \in F^c$ for every $F \in \mathcal{F}$, and therefore $x \in \cap\{F^c \mid F \in \mathcal{F}\}$. In other words, $U^c \subseteq \cap\{F^c | F \in \mathcal{F}\}$.

To show the required equality, we prove $\cap\{F^c | F \in \mathcal{F}\} \subseteq U^c$. To this end, let $x \in \cap\{F^c | F \in \mathcal{F}\}$. This implies that $x \in F^c$ for every $F \in \mathcal{F}$. Hence, $x \notin F$ for every $F \in \mathcal{F}$, and thus we have $x \notin U = \cup\{F \mid F \in \mathcal{F}\}$. In other words, $x \in U^c$, namely $\cap\{F^c | F \in \mathcal{F}\} \subseteq U^c$ as required.

7. In each of the following cases, find the largest ϵ such that S contains an ϵ-neighborhood of x_0.

(i) $x_0 = 3/4 \quad S = \left[\frac{1}{2}, 1\right)$

(ii) $x_0 = 1 \quad S = (0, 2)$

Solution:

(i) The largest ϵ is $\epsilon = 1/4$. Indeed, if $\epsilon = 1/4$, then $(x_0 - \epsilon, x_0 + \epsilon) = (1/2, 1) \subseteq [1/2, 1)$, while if $\epsilon > \frac{1}{4}$, then $x_0 + \epsilon > 1$, therefore $(x_0 - \epsilon, x_0 + \epsilon) \not\subseteq [1/2, 1)$.

(ii) The largest ϵ is $\epsilon = 1$. Indeed if $\epsilon = 1$, then $(x_0 - \epsilon, x_0 + \epsilon) = (0, 2) \subseteq (0, 2)$, while if $\epsilon > 1$, then $x_0 + \epsilon = 1 + \epsilon > 2$, hence $(x_0 - \epsilon, x_0 + \epsilon) \not\subseteq (0, 2)$.

8. Determine if the following sets are open, closed, or neither. Also, determine the set of their interior points, denoted by S^0, the set $(S^c)^0$, and the set $(S^0)^c$.

(i) $S = (-1, 2) \cup [3, \infty)$

(ii) $S = (-\infty, 1) \cup (2, \infty)$

(iii) $S = [-3, -2] \cup [7, 8]$

(iv) $S = \{x \in \mathbb{R} \mid x \in \mathbb{N}\}$

Solution:

(i) The set S is not open, because 3 is not an interior point of S, indeed every ϵ-neighbourhood of 3 contains some point of S^c. The set S is not closed because the point -1 is a limit point but is not contained in S, and a set is closed if and only if it contains all its limit points. The set S^0 is given by $(-1, 2) \cup (3, \infty)$, so $(S^0)^c = (-\infty, -1] \cup [2, 3]$, while $S^c = (-\infty, -1] \cup [2, 3)$, and therefore $(S^c)^0 = (-\infty, -1) \cup (2, 3)$.

(ii) The set S is open because it is union of two open intervals, hence $S^0 = S$, $(S^0)^c = [1, 2]$. Moreover, since $S^c = [1, 2]$, we have that $(S^c)^0 = (1, 2)$.

(iii) The set S is closed because its complement $S^c = (-\infty, -3) \cup (-2, 7) \cup (8, \infty)$ is open, as it is a union of open intervals. Moreover, the set S^0 is given by $(-3, -2) \cup (7, 8)$. Therefore, $(S^0)^c = (-\infty, -3] \cup [-2, 7] \cup [8, \infty)$, while $S^c = (-\infty, -3) \cup (-2, 7) \cup (8, \infty)$ which is open, as it is a union of open intervals, and so $(S^c)^0 = S^c$.

(iv) The set S is not open because for example 1 is not an interior point of S, indeed every ϵ-neighbourhood of 1 contains some point which is in $\mathbb{R}$, but not in $\mathbb{N}$, and hence contains a point of S^c. It is closed because its complement

$$S^c = (-\infty, 1) \cup \bigcup_{n \in \mathbb{N}} (n, n + 1)$$

is a union of open intervals. Moreover, from this we may conclude that $S^c = (S^c)^0$. Finally, since every deleted ϵ-neighbourhood of n contains a real number which is not a natural number, S^0 is empty, and hence $(S^0)^c = \mathbb{R}$.

9. (i) Show that the intersection of finitely many open sets is open.

(ii) Give an example showing that the intersection of infinitely many open sets may fail to be open.

Solution:

(i) Let $U_1, \ldots, U_n$ be open sets. We want to prove that

$$V = \bigcap_{i=1}^{n} U_i$$

is open. Observe that, if V is empty, then it is open, since the emptyset is open. Therefore, without loss of generality, suppose that V is non-empty. If x_0 is a point in V, then $x_0 \in U_i$ for all $i \in \{1, \ldots, n\}$. Since U_i is open, then there exists an ϵ_i-neighbourhood of x_0 contained in U_i for all $i \in \{1, \ldots, n\}$. Let ϵ be $\min\{\epsilon_1, \ldots, \epsilon_n\}$ and observe that $(x_0 - \epsilon, x_0 + \epsilon) \subseteq (x_0 - \epsilon_i, x_0 + \epsilon_i) \subseteq U_i$ for all $i \in \{1, \ldots n\}$. Therefore, $(x_0 - \epsilon, x_0 + \epsilon) \subseteq V$, and so V is open.

(ii) For $n \in \mathbb{N}$, let $U_n = (-1/n, 1/n)$, and observe that

$$\bigcap_{n \in \mathbb{N}} U_n = \bigcap_{n \in \mathbb{N}} (-1/n, 1/n) = \{0\}$$

which is a singleton, and hence closed.

10. For each S write down the set of limit points of S and determine the sets ∂S and $\overline{S}$. Moreover, determine the set of isolated points of S as well as the exterior of S.

(i) $S = \mathbb{Z}$

(ii) $S = \bigcup_{n \in \mathbb{Z}} (n, n+1)$

(iii) $S = \{x \mid x = 1/n \text{ for some } n \in \mathbb{N}\}$

Solution:

(i) The set of limit points of S is the empty set and $\partial S = S$, hence $\overline{S} = S$. The set of isolated points of S coincides with S, and the exterior of S is

$$\bigcup_{n \in \mathbb{Z}} (n, n+1).$$

(ii) The set of limit points of S is

$$\bigcup_{n \in \mathbb{Z}} [n, n+1] = \mathbb{R},$$

and $\partial S = \mathbb{Z}$, hence $\overline{S} = \mathbb{R}$. Moreover, the set of isolated points of S is empty, while the exterior of S is empty because $S^c = \mathbb{Z}$ has no interior points.

(iii) The set of limit points of S is $\{0\}$, and $\partial S = \{0\} \cup S$, hence $\overline{S} = \partial S$. The set of isolated points of S coincides with S and the exterior of S is

$$(-\infty, 0) \cup (1, \infty) \cup \bigcup_{n \in \mathbb{N}} (1/(n+1), 1/n).$$

11. Let S be a nonempty subset of $\mathbb{R}$ such that if $\mathcal{H}$ is any open covering of S, then S has an open covering $\widetilde{\mathcal{H}}$ comprised of finitely many open sets from $\mathcal{H}$. Show that S is closed and bounded.

Solution: We first prove that S is bounded, and then that S is closed.

To this end, fix a point $x_0 \in S$ and consider the following open cover of S

$$\mathcal{H} = \{(x_0 - k, x_0 + k) \mid k \geq 1\},$$

By the hypothesis of the question the set S is covered by finitely many open sets

$$\{H_1 = (x_0 - k_1, x_0 + k_1), \ldots H_n = (x_0 - k_n, x_0 + k_n)\}.$$

Letting $k = \max(k_1, \ldots, k_n)$, we observe that $S \subseteq (x_0 - k, x_0 + k)$. Hence S is bounded since $|x - x_0| < k$ for all $x \in S$, which implies that $|x| \leq k + |x_0|$ for all $x \in S$.

To show that S is closed, we prove that that every limit point of S belongs to S. Let $x \in \mathbb{R}$ be a limit point of S. Suppose, by way of contradiction, that $x \notin S$. For each $s \in S$ set

$$\epsilon_s = |s - x|/2 > 0$$

and let $U_s = (s - \epsilon_s, s + \epsilon_s)$ and observe that $x \notin U_s$ for every $s \in S$, and that $\{U_s \mid s \in S\}$ is an open cover of S. By the hypothesis of the question there is a finite subcover $\{U_{s_1}, \ldots, U_{s_n}\}$ that covers S. Setting $\delta = \min\{|s_i - x|/2 \mid i \in \{1, \ldots, n\}\}$ and observing that $\delta > 0$, we have that the open interval $(x - \delta, x + \delta)$ contains no point of any U_{s_i}, and hence it contains no point of S. This contradicts the assumption that x is a limit point of S, yielding a contradiction to our assumption that S does not contain all its limit points.

12. Let

$$S = \left\{ \frac{(-1)^n}{n} \mid n \in \mathbb{N} \right\},$$

and determine the following, justifying your answer.

(i) The set of interior points of S.

(ii) The set of exterior points of S.

(iii) The set of limit points of S.

(iv) The set of boundary points of S and hence the closure of S.

Solution:

(i) Observe, for each $n \in \mathbb{N}$, and for all positive $\epsilon < (n(n+2))^{-1}$, that

$$((-1)^n n^{-1} - \epsilon, (-1)^n n^{-1} + \epsilon) \setminus \{(n(n+2))^{-1}\} \neq \emptyset$$

since the set $((-1)^n n^{-1} - \epsilon, (-1)^n n^{-1} + \epsilon) \setminus \{(n(n+2))^{-1}\}$ is a union of two non-empty open intervals, indeed this disjoint union of open intervals contains the point $(n+3)/(2n(n+2))$. Moreover, we observe that

$$((-1)^n n^{-1} - \epsilon, (-1)^n n^{-1} + \epsilon) \cap S = \{(-1)^n n^{-1}\}.$$

Thus, S has no interior points.

(ii) Noting that singletons are not open sets, and that the complement S^c of S is

$$(-\infty, -1) \cup \left(\bigcup_{n=0}^{\infty} \left(-\frac{1}{2n+1}, -\frac{1}{2n+3} \right) \right) \cup \{0\} \cup \left(\bigcup_{n=1}^{\infty} \left(\frac{1}{2n+2}, \frac{1}{2n} \right) \right) \cup \left(\frac{1}{2}, \infty \right),$$

a disjoint union of open intervals together with the singleton $\{0\}$, we have that $S^c - \{0\}$ is open. Therefore, the exterior $(S^c)^0$ of S is $S^c - \{0\}$.

(iii) From Part (i), we may conclude that no element of S is a limit point of S. Moreover, if $x < -1$ or $x > 1/2$, setting $\epsilon = \min\{|-1 - x|, |1/2 - x|\}$ we have that

$$(x - \epsilon, x + \epsilon) \cap S = \emptyset,$$

and hence is not a limit point of S. Additionally, if $x \in [-1, 0) - S$, then there exists an $n \in \mathbb{N}$ such that $-1/(2n+1) < x < -1/(2n+3)$ and hence x is not a limit point of S; similarly, if $x \in (0, 1/2] - S$, then there exists an $n \in \mathbb{N}$ such that $1/(2n+2) < x < 1/(2n)$ and hence x is also not a limit point of S. The last case to consider, is if $x = 0$ is a limit point of S, but given an $\epsilon > 0$, there exists an $n \in \mathbb{N}$ such that $1/(2n) < \epsilon$, and hence $(-\epsilon, \epsilon) \cap S \neq \emptyset$. Since this holds for all ϵ, we have that the set of limit points of S is $\{0\}$.

(iv) By Part (i), (ii) and (iii), we may conclude that set of boundary points ∂S of S is $S \cup \{0\}$ and thus the closure is $\overline{S} = S \cup \partial S = S \cup \{0\}$.